

# Nathan Ramon Lau

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<https://nathan-engineering.com>

## EDUCATION

**University of Virginia | School of Engineering and Applied Sciences**

*B.S. in Mechanical Engineering, Minor in Electrical Engineering*

**Cumulative GPA:** 3.80

**Honors:** Dean's List (Academic Year: 2024-25)

**Charlottesville, VA**

*August 2024 – May 2027*

## RELEVANT EXPERIENCE

**BIER Lab at UVA**

*Robotics Researcher*

**Charlottesville, VA**

*August 2025 – Present*

- Retrofitted a robotic fish drone with pressure sensor control algorithms for autonomous navigation through water turbulence
- Designed and fabricated a custom experimental rig, executed controlled testing on the Tunabot to develop a computer vision model that enables real time fish recognition for future autonomous schooling behavior
- Used Python based computer vision software to track fish speed and direction in conjunction with a Hailo processor for real time feedback

**Bio-Inspired Underwater Robotic Submersible Team**

*Lead Engineer and President*

**Charlottesville, VA**

*August 2024 – Present*

- Led the Structures and Safety team of 5 students through ground-up designing and manufacturing targeting the 2026 MATE ROV Competition
- Designed and tested the ROV frame using CAD, FEA and buoyancy calculations to engineer dynamic buoyancy controls for depth adjustment along with a detachable power distribution module
- Organized a team of 20+ students, setting goals, structuring timelines to ensure competition readiness as well as creating bill of materials for the 2025-26 academic year

**Gizmology**

*Project Engineer*

**Charlottesville, VA**

*August 2025 – Present*

- Designed, prototyped, and 3D printed interactive engineering projects for university students, in and around UVA's engineering facilities, showcasing exciting and playful engineering, blending art with technology

**Quadruped Robot**

*Engineer*

**Arlington, VA**

*January 2025 – Present*

- Built and developed a functional quadruped robot dog, combining mechanical design, 3D printing, and electrical assembly with the goal of walking freely
- Used an Arduino and ESP32 to implement robotic motion, utilizing inverse kinematics for leg manipulation using 12 servo motors
- Set up a mock ROS2 interface to create gait paths for lifelike animations as well as wireless remote control

## WORK EXPERIENCE

**University of Virginia Climbing Center**

*Climbing Center Supervisor*

**Charlottesville, VA**

*February 2025 – Present*

- Manage daily operations of UVA's Student Climbing Center ensuring safety and community engagement, overseeing crowds of up to 40 people

**Arlington County Parks & Recreation**

*Gymnastics Coach*

**Arlington, VA**

*June 2023 – August 2025*

- Taught gymnastics and facilitated creative activities for children ages 6 - 10, in groups of 10, changing weekly, for 9 weeks
- Completed safety and inclusion training: workplace safety, injury protocols, fire/hazard response, CPR certified

## TECHNICAL SKILLS & ACTIVITIES

**Society of Hispanic Professional Engineers:** Active member participating in professional development events and attended the 2026 SHPE National Convention

**Design and Programming:** SolidWorks (CSWA Certified), Fusion 360, AutoCAD, Revit, Onshape, Python, C, C++, MATLAB, HTML, JavaScript, Microsoft 360 (Excel, Word, PowerPoint)

**Mechatronics:** 3D Printing, Arduino, ESP32, ROS2, Custom PCB Design, G-Code, Injection molding, CNC machining

**Languages:** Fluent in English and Spanish